

Quiz 4 – Applied OS: Cyber-Defense & Agentic AI

L2: Student Name: _____

Q1		Q2	Total
(3)	(4)	(3)	(10)

Q1 Let's assume the OS is using "**stride scheduling**" for process scheduling. Here we have three processes {A, B, C} with ticket values {A = 100, B = 50, C = 250 tickets}.

a) Assuming a large value like 10,000, determine the stride value of each process. Write the stride value for each process in Row 2 of the table below. (3)

- **Stride(A) = 10,000 / 100 = 100**
- **Stride(B) = 10,000 / 50 = 200**
- **Stride(C) = 10,000 / 250 = 40**

b) For the first three runs only, assume that first A runs, then B, and then C (as shown in the "Who Runs" column). Complete the rest of the table to show how the processes run using stride scheduling. (4)

Stride = 100	Stride = 200	Stride = 40	Who Runs?
Pass(A)	Pass(B)	Pass (C)	
0	0	0	A
100	0	0	B
100	200	0	C
100	200	40	C
100	200	80	C
100	200	120	A
200	200	120	C
(Stop here, as you are required to complete only 7 rows)			...

Q2 The following processes ran via “**lottery scheduling**”.

(3)

```
Process sample_lottery_jobs[] = {
    {1, 5, 5, 60}, // PID 1: 5ms burst, 60 tickets (High Priority)
    {2, 5, 5, 30}, // PID 2: 5ms burst, 30 tickets (Medium Priority)
    {3, 5, 5, 10}  // PID 3: 5ms burst, 10 tickets (Low Priority)
};
```

The output of these processes is shown below:

```
=====
                        RUN 1
=====
Winning Ticket: 28 | Process 1 runs!
Winning Ticket: 31 | Process 1 runs!
Winning Ticket: 81 | Process 2 runs!
Winning Ticket: 36 | Process 1 runs!
Winning Ticket: 3  | Process 1 runs!
Winning Ticket: 78 | Process 2 runs!
Winning Ticket: 3  | Process 1 runs!
Winning Ticket: 10 | Process 2 runs!
Winning Ticket: 4  | Process 2 runs!
Winning Ticket: 27 | Process 2 runs!
Winning Ticket: 1  | Process 3 runs!
Winning Ticket: 7  | Process 3 runs!
Winning Ticket: 9  | Process 3 runs!
Winning Ticket: 3  | Process 3 runs!
Winning Ticket: 8  | Process 3 runs!
```

Notice that in 15 total runs, process 1 runs 5 ×, process 2 runs 5 ×, and process 3 runs 5 ×. Shouldn’t the processes run in the same proportion as their ticket counts {60, 30, 10}? Explain what is happening above.

The processes run exactly 5 times each because they have a finite, short burst time of 5ms. Lottery scheduling only guarantees proportional execution time over long or continuous workloads; for short jobs, higher ticket counts simply mean the job finishes and exits the system sooner.

Here is the step-by-step breakdown of the execution log:

- **Runs 1–7:** All three processes are in the pool (100 total tickets). Process 1 wins the lottery 5 times (winning tickets 28, 31, 36, 3, 3). Because its burst time is only 5ms, Process 1 completes its execution and is removed from the system, taking its 60 tickets with it.
- **Runs 8–10:** Only Process 2 (30 tickets) and Process 3 (10 tickets) remain in the active pool. Process 2 dominates the draws and runs 3 more times (tickets 10, 4, 27). Having already run twice in the first phase (tickets 81, 78), Process 2 reaches its 5ms burst limit and exits the system.
- **Runs 11–15:** Process 3 is the only remaining job. Holding 100% of the remaining tickets, it automatically wins every subsequent draw (tickets 1, 7, 9, 3, 8) until its 5ms burst time is fulfilled.